

Probability

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1 Basics

Definition 1.1. A triple $(\Omega, \Sigma, \mathbb{P})$ where (Ω, Σ) is a measure space and $\mathbb{P}$ is a probability measure (i.e. $\mathbb{P}(\Omega) = 1$) is called a probability space. A measurable function $X : \Omega \rightarrow \mathbb{R}$ is called a random variable.

Examples.

1. "Tossing a coin"

Define $\Omega = \{0, 1\}$, $\Sigma = 2^\Omega = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$ then $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\{0, 1\}) = 1$, $\mathbb{P}(\{0\}) = \mathbb{P}(\{1\}) = \frac{1}{2}$.

$$X : \begin{cases} 0 & \mapsto 0 \\ 1 & \mapsto 1 \end{cases}$$

2. "Roll a die, spell the number, count the number of letters"

Define $\Omega = \{1, \dots, 6\}$, $\Sigma = 2^\Omega$, $\mathbb{P}(\{1\}) = \dots = \mathbb{P}(\{6\}) = \frac{1}{6}$ and extend by additivity.

3. "Tossing a coin twice, take the sum of the outcomes"

Define $\Omega = \{0, 1\} \times \{0, 1\}$, $\Sigma = 2^\Omega$, $\mathbb{P}(\{0, 0\}) = \mathbb{P}(\{0, 1\}) = \mathbb{P}(\{1, 0\}) = \mathbb{P}(\{1, 1\}) = \frac{1}{4}$ then $X((0, 0)) = 0$, $X((0, 1)) = 1$, $X((1, 0)) = 1$, $X((1, 1)) = 2$.

4. "Tossing a coin repeatedly"

Define $\Omega = [0, 1]$, $\Sigma = \mathcal{B}$, $\mathbb{P} = \text{Leb}$. The average of the first 10 coin tosses is

$$X(\omega \in [0, 1]) = \frac{\omega_1 + \dots + \omega_{10}}{10}$$

5. All "named" distributions will come from the Skorokhod Theorem.

Definition 1.2. Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space and X be a random variable then

$$\mu(B) = \mathbb{P}(X^{-1}(B)), \quad B \in \mathcal{B}$$

is called the law of X (it is a probability measure on $(\mathbb{R}, \mathcal{B})$). $F(t) = \mu((-\infty, t])$, $t \in \mathbb{R}$ is called the distribution function of X .

Theorem 1.3. (Properties of F)

1. F is increasing
2. $F(t) \rightarrow 0$ as $t \rightarrow -\infty$, $F(t) \rightarrow 1$ as $t \rightarrow +\infty$
3. F is right-continuous

Proof.

1. Let $s < t$ then

$$F(s) = \mu((-\infty, s]) \leq \mu((-\infty, t]) = F(t)$$

2. Let $t_n \searrow -\infty$,

$$\lim_{n \rightarrow \infty} F(t_n) = \lim_{n \rightarrow \infty} \mu((-\infty, t_n]) = \mu\left(\bigcap_{n \rightarrow \infty} (-\infty, t_n]\right) = \mu(\emptyset) = 0$$

Let $t_n \nearrow +\infty$,

$$\lim_{n \rightarrow \infty} F(t_n) = \lim_{n \rightarrow \infty} \mu((-\infty, t_n]) = \mu\left(\bigcup_{n \rightarrow \infty} (-\infty, t_n]\right) = \mu(\mathbb{R}) = 1$$

3. Let $t_n \searrow t$,

$$\lim_{n \rightarrow \infty} F(t_n) = \lim_{n \rightarrow \infty} \mu((-\infty, t_n]) = \mu\left(\bigcap_{n \rightarrow \infty} (-\infty, t_n]\right) = \mu((-\infty, t]) = F(t)$$

□

Theorem 1.4. (Skorokhod Representation) If $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfies Theorem 1.3 then it is a distribution function of some random variable X .

Idea: $\Omega = [0, 1]$, $\Sigma = \mathcal{B}$, $\mathbb{P} = \text{Leb}$, $X = F^{-1}$. First the condition of X becomes strictly increasing and continuous. $X(\omega) = \inf\{u : F(u) > \omega, \omega \in [0, 1]\}$.

Definition 1.5. X is called a continuous random variable, if there exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$F(x) = \int_{-\infty}^x f(t) dt$$

and f is called the density of X .

Properties of $f(t)$:

1. $\int_{-\infty}^{\infty} f(t) dt = 1$

2. If F is differentiable then $f = F'$.

Definition 1.6. X is called a discrete random variable if there are finite or countable sets $a_1, a_2, \dots \in \mathbb{R}$ and $p_1, p_2, \dots \in [0, 1]$ satisfies $\sum_i p_i = 1$ such that $\forall i, \mathbb{P}(\{\omega : X(\omega) = a_i\}) = p_i$ or equivalently $\mathbb{P}(X = a_i) = p_i$.

Examples. 1. Continuous distributions:

(a) (Normal distribution) $N(0, 1)$ where

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

for $N(\mu, \sigma^2)$ we have

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(b) (Exponential distribution) $\exp(\mu)$ where

$$f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{x}{\mu}} & x > 0 \\ 0 & x < 0 \end{cases}$$

(c) (Uniform Distribution) $U[a, b]$ where

$$f(x) = \frac{1}{b-a}$$

2. Discrete distributions:

(a) (Bernoulli)(p) where

$$\begin{cases} a_1 = 0 & p_1 = 1-p \\ a_2 = 1 & p_2 = p \end{cases}$$

(b) (Binomial)(p, N) where

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

(c) (Poisson)(λ) where

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

where $k = 0, 1, 2, \dots$

3. Neither continuous nor discrete: $F = \text{Cantor function}$.

Definition 1.7. The expectation of X is defined by

$$\mathbb{E}[X] = \int_{\Omega} X dP$$

if X is Lebesgue integrable. If $X \geq 0$ and not integrable then $\mathbb{E}(X) = +\infty$.

Theorem 1.8. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be integrable on $(\mathbb{R}, \mathcal{B}, \mu)$. Then

$$\mathbb{E}[h(X)] = \int_{\Omega} h(X) dP = \int_{\mathbb{R}} h(x) d\mu$$

If X is continuous random variable then

$$\mathbb{E}[h(x)] = \int_{\mathbb{R}} h(x)p(x) dx$$

If X is discrete random variable then

$$\mathbb{E}[h(x)] = \sum_i h(a_i) \mathbb{P}(X = a_i)$$

Proof. Let $h = \mathbb{1}_B$, with $B \in \mathcal{B}$. Then

$$\int_{\Omega} h(x) dP = \int_{\Omega} \mathbb{1}_B dP = \mathbb{P}(\{\omega : X(\omega) \in B\})$$

while

$$\int_{\mathbb{R}} \mathbb{1}_B(x) d\mu = \mu(B)$$

thus it works for simple functions. Similarly, by measure theory we are able to extend the theorem to all functions. $\square$

Theorem 1.9. (Markov& Chebychev inequality) Let X be non-negative random variable. Then $\forall c \geq 0$,

$$\mathbb{P}(\omega : X(\omega) \geq c) \leq \frac{1}{c} \mathbb{E}[X]$$

Proof. Let $A = \{\omega : X(\omega) \geq c\}$ then

$$\begin{aligned}\mathbb{E}[X] &= \int_{\Omega} X(\omega) \, dP \geq \int_A X(\omega) \, dP \geq \int_A c \, dP \\ &= c \int_A dP = c\mathbb{P}(A) = c\mathbb{P}(\{\omega : X(\omega) \geq c\})\end{aligned}$$

□

Definition 1.10. A random variable X is called square integrable if $E[X^2] < \infty$. Define $Var(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$ is the variance of X .

If X is square-integrable then it is integrable,

Proof. By Cauchy-schwarz,

$$\mathbb{E}[|X|] \cdot 1 \leq \sqrt{\mathbb{E}[X^2]} \cdot \sqrt{1^2} < \infty$$

□

Note that this does not hold for general L^1 and L^2 since $L^2 \not\subset L^1$ but work well here since $L^2_{loc} \subset L^1_{loc}$.

2 Independence

Definition 2.1. Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space with finitely many σ -algebras $\Sigma_1, \Sigma_2, \dots, \Sigma_m \subset \Sigma$ are independent if

$$\mathbb{P}(A_1 \cap \dots \cap A_m) = \mathbb{P}(A_1) \dots \mathbb{P}(A_m)$$

for any $A_1 \in \Sigma_1, \dots, A_m \in \Sigma_m$. Countably many σ -algebras $\Sigma_1, \Sigma_2, \dots$ if any finite subcollection of them is independent. The events $(E_i) \in \Sigma$ are independent if their generated σ -algebras are independent. The random variables (X_i) are independent if their generated σ -algebras are independent.

For $E \in \Sigma$, $\sigma(E) = \{\emptyset, \Omega, E, \Omega \setminus E\}$. For a random variable X , $\sigma(X) = \sigma\{X^{-1}(B), B \in \mathcal{B}\}$.

Example. A, B, C are independent if

$$\sigma(A) = \{\emptyset, \Omega, A, \Omega \setminus A\}$$

$$\sigma(B) = \{\emptyset, \Omega, B, \Omega \setminus B\}$$

$$\sigma(C) = \{\emptyset, \Omega, C, \Omega \setminus C\}$$

are independent. Enough to check pairs and triples satisfy the definition of independence.

Example. "Toss a coin, then throw a die" $\Omega = \{0, 1\} \times \{1, \dots, 6\}$, $\Sigma = 2^\Omega$, $\mathbb{P}(\{(a, b)\}) = \frac{1}{12}$ and

$$X(a, b) = a$$

$$Y(a, b) = b$$

Are X and Y independent?

$$\sigma(X) = \{\emptyset, \Omega, X(0, b), X(1, b)\}$$

$$\sigma(Y) = \{\emptyset, \Omega, Y(a, 1), Y(a, 2), Y(a, 3), Y(a, 4), Y(a, 5), Y(a, 6)\}$$

and we are able to check

$$\mathbb{P}((a_1, b) \cap (a, b_1)) = \mathbb{P}(a_1, b) \mathbb{P}(a, b_1)$$

Definition 2.2. A collection of sets $\mathcal{I}$ is a π -system if

1. $\emptyset \in \mathcal{I}$
2. if $A, B \in \mathcal{I}$ then $A \cap B \in \mathcal{I}$

Examples.

1. $\mathbb{R} : \{(-\infty, t], t \in \mathbb{R}, \emptyset\}$
2. $(\Omega, \Sigma, \mathbb{P}), X : \{\{X \leq t\}, t \in \mathbb{R}, \emptyset\}$

Theorem 2.3. If $\mathcal{I}$ and $\mathcal{J}$ are two independent π -systems, then their generated σ -algebras are independent. This is equivalent to

$$\mathbb{P}(I \cap J) = \mathbb{P}(I) \mathbb{P}(J)$$

for all $I \in \mathcal{I}$ and $J \in \mathcal{J}$.

If X and Y are continuous random variable, $\sigma(x)$ is generated by $\{X \leq x\}$ and $\{Y \leq y\}$ then

$$X \perp Y \Leftrightarrow \mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x) \mathbb{P}(Y \leq y)$$

for all x, y . If X and Y are discrete with values $(a_i), (b_i)$ then

$$\sigma(X) = \{x = a_i\}, \quad \sigma(Y) = \{y = b_i\}$$

then

$$X \perp Y \Leftrightarrow \mathbb{P}(X = a_i, Y = b_j) = \mathbb{P}(X = a_i) \mathbb{P}(Y = b_j)$$

Theorem 2.4. Let $\mathcal{I}$ be a π -system and $\sigma(\mathcal{I})$ its generated σ -algebra. Suppose μ_1 and μ_2 are two measures on $\sigma(\mathcal{I})$ such that

- (1) $\mu_1(I) = \mu_2(I)$ for any $I \in \mathcal{I}$.
- (2) $\mu_1(\Omega) = \mu_2(\Omega) < \infty$

Then $\mu_1 = \mu_2$ on $\sigma(\mathcal{I})$.

Proof. (Theorem 2.3) We need $\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$ for any $A \in \sigma(\mathcal{I}), B \in \sigma(\mathcal{J})$.

(Step 1) Fix $J \in \mathcal{J}$, let μ_1, μ_2 be two measures on $\sigma(\mathcal{I})$ where

$$\mu_1(A) = \mathbb{P}(A \cap J), \quad \mu_2(A) = \mathbb{P}(A) = \mathbb{P}(A) \mathbb{P}(J)$$

where $A \in \sigma(\mathcal{I})$. (1) from Theorem 2.4 is satisfied and (2) from Theorem 2.4, $\mu_1(\Omega) = \mathbb{P}(J) = \mathbb{P}(\Omega) \mathbb{P}(J) = \mu_2(\Omega)$.

Thus, by theorem,

$$\mu_1(A) = \mu_2(A), \forall A \implies \mathbb{P}(A \cap J) = \mathbb{P}(A) \mathbb{P}(J)$$

(Step 2) Fix $A \in \sigma(\mathcal{I})$ Let

$$\mu_1(B) = \mathbb{P}(A \cap B), \quad \mu_2(B) = \mathbb{P}(A)\mathbb{P}(B)$$

where $B \in \sigma(\mathcal{J})$. Again, by Theorem 2.4 we have $\mu_1(B) = \mu_2(B), \forall B$. Thus, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. $\square$

Given F_1 and F_2 , construct X, Y such that

- (a) F_1 is the distribution function of X . F_2 is the distribution function of Y .
- (b) X and Y are independent.

Let $\tilde{X}$ be a random variable with distribution function F_1 on $(\Omega_1, \Sigma_1, \mathbb{P}_1)$, $\tilde{Y}$ be a random variable with distribution function F_2 on $(\Omega_2, \Sigma_2, \mathbb{P}_2)$. Now $\Omega = \Omega_1 \times \Omega_2$, $\Sigma = \Sigma_1 \times \Sigma_2$ and $\mathbb{P} = \mathbb{P}_1 \otimes \mathbb{P}_2$. Define

$$X(\omega_1, \omega_2) = \tilde{X}(\omega_1), \quad Y(\omega_1, \omega_2) = \tilde{Y}(\omega_2)$$

$$F_X(x) = (\mathbb{P}_1 \otimes \mathbb{P}_2)(X \leq x) = \mathbb{P}_1(\tilde{X}_1 \leq x)\mathbb{P}_2(\Omega_2) = F_1(x)$$

Similarly,

$$F_Y(y) = F_2(y)$$

To show independence,

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(X \leq x, Y \leq y) = \mathbb{P}_1(\tilde{X} \leq x)\mathbb{P}_2(\tilde{Y} \leq y)$$

$$= (\mathbb{P}_1 \otimes \mathbb{P}_2)(X \leq x) \cdot (\mathbb{P}_1 \otimes \mathbb{P}_2)(Y \leq y)$$

Given (F_n) , construct (X_n) such that

- (a) (F_n) is the distribution function of $X_n, \forall n$.
- (b) (X_n) are independent.

"Trick model" construction. $([0, 1], \mathcal{B}, \text{Leb})$ say $[0, 1] \ni \omega = 0.\omega_1\omega_2 \cdots \mapsto \omega_n$. ω_n has Bernoulli($\frac{1}{2}$) and (ω_n) are independent.

$$\text{Leb}(\omega_1 = 0, \omega_2 = 1) = \text{Leb}(\omega_1 = 0)\text{Leb}(\omega_2 = 1)$$

Let

$$X_1(\omega) = 0.\omega_1\omega_5 \cdots$$

$$X_2(\omega) = 0.\omega_5\omega_{102} \cdots$$

$$X_3(\omega) = 0.\omega_4\omega_{180} \cdots$$

are uniform and independent. We know now have to construct (U_n) such that each U_n is uniform and (U_n) are independent. If the inverse exists, $X_n = F_n^{-1}(U_n) : F_{X_n}(t) = \text{Leb}(X_n \leq t) = \text{Leb}(F_n^{-1}(U_n) \leq t) = \text{Leb}(U_n \leq F_n(t)) = F_{U_n}(F_n(t)) = F_n(t)$.

Theorem 2.5. Let X and Y be independent variables

1. If X and Y are integrable then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.
2. If X and Y are square-integrable then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Proof.

- (a) First, assume that $X = \mathbb{1}_A$ and $Y = \mathbb{1}_B$ and X and Y are independent or $\sigma(X)$ and $\sigma(Y)$ are independent then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. $\mathbb{E}(XY) = \mathbb{E}(\mathbb{1}_A \mathbb{1}_B) = \mathbb{E}(\mathbb{1}_{A \cap B}) = \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = \mathbb{E}(X)\mathbb{E}(Y)$. Then use to this to derive simple functions, non-negative functions and general functions.
- (b) $\text{Var}(X + Y) = \mathbb{E}((X + Y)^2) - (\mathbb{E}(X + Y))^2 = \mathbb{E}(X^2 + 2XY + Y^2) - (\mathbb{E}(X) + \mathbb{E}(Y))^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 + \mathbb{E}(Y^2) - (\mathbb{E}(Y))^2 + 2[\mathbb{E}(XY) - \mathbb{E}(Y)\mathbb{E}(X)] = \text{Var}(X) + \text{Var}(Y)$.

$\square$

Given $X : F_x$ and $Y : F_y$

Definition 2.6. Let X and Y be random variables defined on the same probability space. The joint law of X and Y is a measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$ defined by $\mu(\mathcal{B}) = \mathbb{P}(\{\omega \in \Omega : (X(\omega), Y(\omega)) \in \mathcal{B}\})$. The joint distribution function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$F(x, y) = \mu((-\infty, x] \times (-\infty, y]) = \mathbb{P}(X \leq x, Y \leq y), \quad x, y \in \mathbb{R}$$

Theorem 2.7. (a) If X and Y are independent then $F(x, y) = F_X(x)F_Y(y)$.

- (b) If X and Y are independent and have densities f and g respectively then μ has density with respect to Leb^2 given by $(x, y) \mapsto f(x)g(y)$.
- 1. If X and Y are independent and have densities f and g respectively then $X + Y$ has density $f * g$ (convolution of f and g) which is defined by

$$f * g(t) = \int_{-\infty}^{\infty} f(s)g(t - s) ds$$

Proof.

(b) We need $\mu(B) = \int_B f(u)g(v) \, dudv$, $\forall B \in \mathcal{B}(\mathbb{R}^2)$. It is enough to check for $B = (-\infty, x] \times (-\infty, y]$.

$$\begin{aligned} \int_{(-\infty, x] \times (-\infty, y]} f(u)g(v) \, dudv &= \int_{-\infty}^x f(u) \, du \int_{-\infty}^y g(v) \, dv \\ &= F_X(x)F_Y(y) = F(x, y) = \mu((-\infty, x] \times (-\infty, y]) \end{aligned}$$

(c) $F_{X+Y}(t) = \mathbb{P}(X + Y \leq t) = \mu(\{(x, y) : x + y \leq t\})$

$$= \int \int f(x)g(y) \, dxdy = \int_{-\infty}^{\infty} f(x) \int_{-\infty}^t g(y) \, dxdy = \int_{-\infty}^{\infty} f(x) \int_{-\infty}^t g(v-x) \, dvdx = \int_{-\infty}^t \int_{-\infty}^{\infty} f(x)g(v-x) \, dxdv$$

differentiation yields the result required. $\square$

Infinite and eventual occurrence of event. Let $\{E_n\}$ be sequence of events and

$$\{E_n \text{ i.o.}\} = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n$$

be E_n occurs infinitely often and

$$\{E_n \text{ ev.}\} = \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} E_n$$

be E_n occurs eventually.

Example. "Toss a coin repeatedly" and define $E_n = \{\text{heads at time } n\}$ then

$$\mathbb{P}(E_n \text{ i.o.}) = 1, \quad \mathbb{P}(E_n \text{ ev.}) = 1$$

$$\begin{aligned} \{X_n \rightarrow X\} &= \{\forall k \in \mathbb{N}, \exists N, \forall n \geq N, |X_n - X| < \frac{1}{k}\} = \\ &= \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{|X_n - X| < \frac{1}{k}\} = \bigcap_{k=1}^{\infty} \{ |X_n - X| < \frac{1}{k} \text{ ev.} \} \end{aligned}$$

Note

$$\{E_n \text{ ev.}\}^c = \left(\bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n \right)^c = \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} E_n^c = \{E_n^c \text{ i.o.}\}$$

Lemma 2.8. (Borel-Cantelli Lemma 1) If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$ then $\mathbb{P}(E_n \text{ i.o.}) = 0$.

Proof.

$$\mathbb{P}(E_n \text{ i.o.}) = \mathbb{P}\left(\bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n\right) = \lim_{N \rightarrow \infty} \mathbb{P}\left(\bigcup_{n \geq N} E_n\right) = 0$$

$\square$

Lemma 2.9. (Borel-Cantelli Lemma 2) If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \infty$, (E_n) are independent then $\mathbb{P}(E_n \text{ i.o.}) = 1$

Proof. We want to show

$$\mathbb{P}\left(\bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n\right) = 1$$

i.e.,

$$\mathbb{P}\left(\bigcup_{N=1}^{\infty} \bigcap_{n \geq N} E_n^c\right) = 0$$

i.e.,

$$\lim_{N \rightarrow \infty} \mathbb{P}\left(\bigcap_{n \geq N} E_n^c\right) = 0$$

$$\begin{aligned} \mathbb{P}\left(\bigcap_{n \geq N} E_n^c\right) &= \lim_{M \rightarrow \infty} \mathbb{P}\left(\bigcap_{n \geq N}^M E_n^c\right) = \lim_{M \rightarrow \infty} \prod_{n=N}^M \mathbb{P}(E_n^c) \leq \lim_{M \rightarrow \infty} \prod_{n=N}^M e^{-\mathbb{P}(E_n)} \\ &= e^{-\lim_{M \rightarrow \infty} \sum_{n=N}^M \mathbb{P}(E_n)} = 0 \end{aligned}$$

$\square$

Example. (X_n) are independent exponential random variables then

$$\limsup \frac{X_n}{\varphi(n)} = 1$$

To prove $\limsup a_n = a$ it suffices to show that $\forall b > a$, $a_n > b$ only finitely often or $\forall b < a$, $a_n > b$ infinitely happen. Take any b ,

$$\mathbb{P}\left(\frac{X_n}{\varphi(n)} > b \text{ i.o.}\right) = \begin{cases} 0 & b > 1 \\ 1 & b < 1 \end{cases}$$

$$\begin{aligned}\sum_{n=1}^{\infty} \mathbb{P}\left(\frac{X_n}{\varphi(n)} > b\right) &= \sum_{n=1}^{\infty} \mathbb{P}(X_n > \varphi(n)b) \\ &= \sum_{n=1}^{\infty} e^{-b\varphi(n)}\end{aligned}$$

If we pick $\varphi(n) = \log n$ then

$$\Rightarrow \sum_{n=1}^{\infty} e^{-b \log n} = \sum_{n=1}^{\infty} \frac{1}{n^b}$$

where

$$\sum_{n=1}^{\infty} \frac{1}{n^b} = \begin{cases} < \infty & b > 1 \\ = \infty & b < 1 \end{cases}$$

Theorem 2.10. (Bernstein's inequality) Let (X_n) be independent and such that $\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2} \forall n$ then

$$\mathbb{P}\left(\left|\sum_{i=1}^n a_i X_i\right| > t\right) \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2}\right)$$

for any $a_1, \dots, a_n$ and any $t > 0$.

Proof. It suffices to prove take $\lambda > 0$

$$\mathbb{P}\left(\exp\left(\lambda \sum_{i=1}^n a_i X_i\right) \geq e^{\lambda t}\right) \leq e^{-\lambda t} \mathbb{E}\left(\exp\left(\lambda \sum_{i=1}^n a_i X_i\right)\right) = e^{-\lambda t} \prod_{i=1}^n \frac{e^{\lambda a_i} + e^{-\lambda a_i}}{2}$$

Use the result that $\cosh x \leq e^{x^2/2}$ then

$$\begin{aligned}-\lambda t \prod_{i=1}^n \cosh(\lambda a_i) &\leq e^{-\lambda t} \prod_{i=1}^n e^{\lambda^2 a_i^2/2} = e^{-\lambda t + \lambda^2 C/2} \\ &\leq \exp\left(-t \frac{t}{C} + \frac{1}{2} \left(\frac{t}{C}\right)^2\right) = \exp\left(-\frac{t^2}{2C}\right)\end{aligned}$$

where $C = \sum_{i=1}^n a_i^2$ as required. $\square$

Theorem 2.11. (Strong law of large numbers for " ± 1 with probability $\frac{1}{2}$ " random variables) Let (X_n) be independent and such that $\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2} \forall n$ then

$$\frac{X_1 + \dots + X_n}{n} \rightarrow 0 \quad a.s.$$

Proof. We want to prove

$$\mathbb{P}\left(\left|\frac{X_1 + \dots + X_n}{n} - 0\right| < \frac{1}{k} \text{ e.v.}\right) = 1$$

i.e.,

$$\mathbb{P}\left(\left|\frac{X_1 + \dots + X_n}{n}\right| \geq \frac{1}{k} \text{ i.o.}\right) = 0$$

Take $a_1 = \dots = a_n = \frac{1}{n}$ then by Bernstein's inequality

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left|\sum_{i=1}^n a_i X_i\right| \geq \frac{1}{k}\right) \leq \sum_{n=1}^{\infty} 2 \exp\left(-\frac{\left(\frac{1}{k}\right)^2}{2 \cdot \frac{n}{n^2}}\right) = 2 \sum_{n=1}^{\infty} \exp\left(-\frac{n}{2k^2}\right) < \infty$$

then by Borel-Cantelli Lemma 1 we have

$$\mathbb{P}\left(\left|\frac{X_1 + \dots + X_n}{n}\right| \geq \frac{1}{k} \text{ i.o.}\right) = 0$$

as required. $\square$

Definition 2.12. Given (X_n) Let $\tau_n = \sigma(X_{n+1}, X_{n+2}, \dots)$ be nth term of σ -algebra. τ_n does not depend on $(X_1, X_2, \dots, X_n)$ define

$$\tau = \bigcap_{n=1}^{\infty} \tau_n$$

is called the tail of σ -algebra. And $E \subset \tau$ is called a tail event (event that doesn't rely on any finite number of (X_n) s).

Examples.

- (1) $\{X_n \rightarrow a\} = \{X_{n+m} \rightarrow a\} \in \tau_m \forall m$ which means $E \in \tau$.
- (2) $\{\lim_{n \rightarrow \infty} X_n \text{ exists}\}$. This is a tail event.
- (3) $\{\sum_{n=1}^{\infty} X_n < \infty\}$. This is a tail event.

- (4) $\{\{\sum_{n=1}^{\infty} X_n > 0\}\}$. This is not a tail event.
- (5) $\{\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_n}{n} \text{ exists}\}$. This is tail event.
- (6) $\{\sup_{n \in \mathbb{N}} X_n > 0\}$. This is not tail event.

Theorem 2.13. (Kolmogorov's 0–1 law). Let (X_n) be a sequence of independent random variable then $\mathbb{P}(E) \in \{0, 1\}$ for any $E \in \tau$.

Proof. Let $\sigma_n = \sigma(X_1, \dots, X_n)$ then

$$\begin{aligned} \sigma_n &\perp \tau_n \quad \forall n \\ \implies \sigma_n &\perp \tau \quad \forall n \end{aligned}$$

Let $\sigma_n = \sigma(X_1, X_2, \dots)$ is generated by the π -system $\sigma_1 \cup \sigma_2 \cup \dots$

$$\begin{aligned} \implies \sigma_{\infty} &\perp \tau \\ \implies \sigma_{\infty} &\perp \tau \implies \tau \perp \tau \end{aligned}$$

Take $E \in \tau : \mathbb{P}(E \cap E) = \mathbb{P}(E)^2$ which is true if and only if $\mathbb{P}(E) \in \{0, 1\}$. □

3 Weak Convergence

Definition 3.1. (X_n) with laws (μ_n) converge weakly or in distribution to X with the law μ we write $X_n \Rightarrow X$ or $\mu_n \Rightarrow \mu$ if $F_n(t) \rightarrow F(t)$ for all t where F is continuous.

Remark. If we don't remove discontinuity points from the definition then for example $\frac{1}{n} \rightarrow 0$.

Theorem 3.2. (a.s. vs weak)

- (1) If $X_n \rightarrow X$ almost surely then $X_n \Rightarrow X$.
- (2) If $\mu_n \Rightarrow \mu$ then there are $(X_n), X$ defined on $([0, 1], \mathcal{B}, \text{Leb})$ with laws $(\mu_n), (\mu)$ respectively such that $X_n \rightarrow X$ a.s.

Proof. (2) $X_n(\omega) = \inf\{u : F_n(u) > \omega\}$ and $X(\omega) = \inf\{u : (u) > \omega\}$ where $\omega \in [0, 1]$. Let

$$B = \{\omega \in [0, 1] : \exists s < t \text{ s.t. } F(s) = F(t) = \omega\}$$

Claim 1: B is at most countable and has measure zero since each non-overlapping interval contains rational numbers which are countable and $\text{Leb}(B) = 0$. Claim 2: discontinuity points of F is at most countable since each non-overlapping interval contains rational numbers which are countable. Then we'll show that $X_n(\omega) \rightarrow X(\omega)$ for all $\omega \in [0, 1] \setminus B$. Let $\omega \in [0, 1] \setminus B$ and given $\varepsilon > 0$, let $\delta \leq \varepsilon$ be such that $x \pm \delta$ are continuous points of F then

$$F(x - \delta) < \omega < F(x + \delta)$$

and by weak convergence $\exists N, \forall n \geq N$

$$\begin{aligned} F_n(x - \delta) &< \omega < F_n(x + \delta) \\ \implies |X_n(\omega) - X(\omega)| &< \delta \leq \varepsilon \end{aligned}$$

□

Proof. (1) Suppose $X_n \rightarrow X$ a.s. Let h be continuous and bounded then it's basically the same to Theorem 3.3. □

Theorem 3.3. (Equivalent definition of weak convergence 1) $\mu_n \Rightarrow \mu$ if and only if

$$\int h d\mu_n \rightarrow \int h d\mu$$

for any $h : \mathbb{R} \rightarrow \mathbb{R}$ which is continuous and bounded.

Proof. ($\implies$) Suppose $\mu_n \Rightarrow \mu$ choose (X_n) as in Theorem 3.2(2). Let h be continuous and bounded then it suffices to show

$$\int_{\mathbb{R}} h d\mu_n = \mathbb{E}(h(X_n)) \rightarrow \mathbb{E}(h(X)) = \int_{\mathbb{R}} h(X_n) d\mu$$

since $X_n \rightarrow X$ a.s. then $h(X_n) \rightarrow h(X)$ a.s. since h is continuous and $h(X_n)$ is bounded so by dominated convergence theorem we are done.

($\impliedby$) Let t be a continuity point of F we will show that

$$F(t) \leq \liminf_{n \rightarrow \infty} F_n(t) \leq \limsup_{n \rightarrow \infty} F_n(t) \leq F(t)$$

To do so let

$$h = \begin{cases} 1 & t \leq x - \delta \\ 0 & t \geq x \\ \in [0, 1] & \text{otherwise} \end{cases}$$

$$F_n(t) = \int \mathbb{1}_{(-\infty, t]} d\mu_n \geq \int h d\mu_n$$

and

$$\liminf_{n \rightarrow \infty} F_n(t) \geq \lim_{n \rightarrow \infty} \int h d\mu_n = \int h d\mu \geq \int \mathbb{1}_{(-\infty, t-\delta)} d\mu_n = F(t - \delta)$$

and the part for $\limsup$ is quite similar. □

Definition 3.4. $h : \mathbb{R} \rightarrow \mathbb{R}$ is a C_0^2 test function if h'' exists and is continuous and it is compactly supported.

Definition 3.5. (μ_n) is tight if $\forall \varepsilon > 0, \exists M$ such that $\mu_n(-M, M) \geq 1 - \varepsilon \forall n$.

Examples.

1. Finite collection of measures is always tight.
2. $\mu_n = \delta_n$ then $\delta_1([-M, M]) = 1 \geq 1 - \varepsilon$.
3. $N(a_n, 1)$ is tight if a_n is bounded.

Lemma 3.6. If

$$\int_{\mathbb{R}} h \, d\mu_n \rightarrow \int_{\mathbb{R}} h \, d\mu$$

for any $h : \mathbb{R} \rightarrow \mathbb{R}$ which is a C_0^2 test function then (μ_n) is tight.

Proof. Let $\varepsilon > 0$, choosing M so that $\mu([-M, M]) \geq 1 - \frac{\varepsilon}{2}$ then $\exists N$ such that $\forall n \geq N$

$$\begin{aligned} \mu_n([-M-1, M+1]) &= \int_{\mathbb{R}} \mathbb{1}_{[-M-1, M+1]} \, d\mu_n \geq \int_{\mathbb{R}} h \, d\mu_n \geq \int_{\mathbb{R}} h \, d\mu - \frac{\varepsilon}{2} \\ &\geq \int_{\mathbb{R}} \mathbb{1}_{[-M, M]} \, d\mu - \frac{\varepsilon}{2} = \mu([-M, M]) - \frac{\varepsilon}{2} \geq 1 - \frac{\varepsilon}{2} - \frac{\varepsilon}{2} = 1 - \varepsilon \end{aligned}$$

□

Theorem 3.7. (Equivalent definition of weak convergence 2) $\mu_n \Rightarrow \mu$ if and only if

$$\int_{\mathbb{R}} h \, d\mu_n \rightarrow \int_{\mathbb{R}} h \, d\mu$$

for any $h : \mathbb{R} \rightarrow \mathbb{R}$ which is a C_0^2 test function.

Proof. ($\Rightarrow$) Follows immediately by Theorem 3.3.

($\Leftarrow$) Let x be a continuity point of F . We will show $F(x) \leq \liminf F_n(x) \leq \limsup F_n(x) \leq F(x)$ as this would imply the limit $\lim_{n \rightarrow \infty} F_n(x) = F(x)$. Let $\delta, \varepsilon > 0$, by Lemma 3.6, $\exists M$ such that $\mu_n([-M, M]) \geq 1 - \varepsilon \, \forall n$.

$$F_n(x) = \int_{\mathbb{R}} \mathbb{1}_{(-\infty, x]} \, d\mu_n + \mu_n((-\infty, -M]) \leq \varepsilon + \int_{\mathbb{R}} h \, d\mu_n$$

$$\limsup F_n(x) \leq \varepsilon + \int_{\mathbb{R}} h \, d\mu \leq \varepsilon + \int_{\mathbb{R}} \mathbb{1}_{(-\infty, x+\delta]} \, d\mu = \varepsilon + F(x+\delta)$$

Take $\varepsilon, \delta \rightarrow 0$ gives $\limsup F_n(x) = F(x)$.

□

Definition 3.8. $\varphi : \mathbb{R} \rightarrow \mathbb{C}$, $\varphi(t) = \mathbb{E}(e^{itx})$ is the characteristic function of X . Define $\hat{\mu} : \mathbb{R} \rightarrow \mathbb{C}$,

$$\hat{u}(t) = \int_{\mathbb{R}} e^{itx} \, d\mu$$

is called the Fourier transform of μ . If $f : \mathbb{R} \rightarrow \mathbb{R}$ is an integrable function then

$$\hat{f}(x) = \int_{\mathbb{R}} f(x) e^{itx} \, dx$$

is the Fourier Transform of f .

Theorem 3.9. (Properties of a characteristic function)

- (1) $\varphi(0) = 1$
- (2) $\varphi_{\lambda X}(t) = \varphi_X(\lambda t)$
- (3) If X and Y are independent then
- (4) φ is continuous.

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$$

Examples.

1. $\mathbb{P}(X = \pm 1) = \frac{1}{2}$ then

$$\varphi(t) = \frac{1}{2}e^{it} + \frac{1}{2}e^{-it} = \cos t$$

2. Uniform distribution on $[-1, 1]$ then

$$\varphi(t) = \frac{1}{2} \int_{-1}^1 e^{itx} \, dx = \frac{\sin t}{t}$$

3. Cauchy distribution $f(x) = \frac{1}{\pi(1+x^2)}$ then

$$\varphi(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{itx}}{1+x^2} \, dx = e^{-|t|}$$

4. $N(0, 1)$ has characteristic equation $\varphi(t) = e^{-t^2/2}$. Proof is omitted here.

Theorem 3.10. (Decay of Fourier Transform)

- (1) Let $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $h \in C^m$ and $h^{(i)}$ are integrable, $h^{(i)}(x) \rightarrow 0$ as $|x| \rightarrow \infty$ for $i \in [0, m]$ then $\exists c$ such that

$$|\hat{h}(t)| \leq \frac{c}{t^m}$$

for all $t \neq 0$.

(2) Let h be a C^2 test function. Then $\exists c$ such that $|\hat{h}(t)| \leq \frac{c}{t^2} \forall t \neq 0$ and in particular $\hat{h}$ is integrable.

Proof. By definition,

$$\begin{aligned} |\hat{h}(t)| &= \left| \int_{-\infty}^{\infty} h(x) e^{itx} dx \right| \\ &= \left| h(x) \frac{e^{itx}}{it} \Big|_{-\infty}^{+\infty} - \frac{1}{it} \int_{-\infty}^{\infty} h'(x) e^{itx} dx \right| \\ &= \left| \frac{1}{t} \int_{-\infty}^{\infty} h'(x) e^{itx} dx \right| \\ &= \left| \frac{1}{t} \int_{-\infty}^{\infty} h'(x) e^{itx} dx \right| \\ &= \left| \frac{1}{t^m} \int_{-\infty}^{\infty} h^{(m)}(x) e^{itx} dx \right| \\ &\leq \frac{C}{|t|^m} \end{aligned}$$

□

Theorem 3.11. (Parseval-Plancherel) For any C^2 test function h and any random variable with law μ and characteristic function φ then

$$\int_{\mathbb{R}} h d\mu = \frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{\hat{h}(t)} \varphi(t) dt$$

Proof. The RHS of the equation is equivalent to

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{\hat{h}(t)} \varphi(t) dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} h(x) e^{-itx} dx \right) \left(\int_{\mathbb{R}} e^{ity} d\mu \right) \\ &= \frac{1}{2\pi} \lim_{T \rightarrow \infty} \int_{-T}^T \left(\int_{-\infty}^{\infty} h(x) e^{-itx} dx \right) \left(\int_{\mathbb{R}} e^{ity} d\mu \right) \\ &= \frac{1}{2\pi} \lim_{T \rightarrow \infty} \int_{\mathbb{R}} \int_{-\infty}^{\infty} h(x) \int_{-T}^T e^{it(y-x)} dt dx d\mu(y) \\ &= \frac{1}{\pi} \lim_{T \rightarrow \infty} \int_{\mathbb{R}} \int_{-\infty}^{\infty} h(x) \frac{e^{it(y-x)}}{2i(y-x)} \Big|_{-T}^T dx d\mu(y) \\ &= \frac{1}{\pi} \lim_{T \rightarrow \infty} \int_{\mathbb{R}} \int_{-\infty}^{\infty} h(x) \frac{\sin(T(y-x))}{y-x} dx d\mu(y) \\ &= \lim_{T \rightarrow \infty} \int_{\mathbb{R}} \frac{1}{\pi} \int_{-\infty}^{\infty} h(x) \frac{\sin(T(x-y))}{x-y} dx d\mu \end{aligned}$$

Let

$$h^T(y) = \frac{1}{\pi} \int_{-\infty}^{\infty} h(x) \frac{\sin(T(x-y))}{x-y} dx$$

We need $h^T(y) \rightarrow h(y)$ and h^T is uniformly bounded.

$$\begin{aligned} h_+^T(y) &= \frac{1}{\pi} \int_y^{\infty} h(x) \frac{\sin(T(x-y))}{x-y} dx = \frac{1}{\pi} h(x) \int_y^x \frac{\sin(T(u-y))}{u-y} du \Big|_y^{\infty} - \frac{1}{\pi} \int_y^{\infty} h'(x) \frac{\sin(T(u-y))}{u-y} du dx \\ &= -\frac{1}{\pi} \int_y^{\infty} h'(x) \int_0^{T(x-y)} \frac{\sin v}{v} dv dx = -\frac{1}{2} \int_y^{\infty} h'(x) dx = \frac{h(y)}{2} \end{aligned}$$

□

Theorem 3.12. $X_n \Rightarrow X$ if and only if $\varphi_n(t) \rightarrow \varphi(t) \forall t$.

Proof. ($\Rightarrow$) By definition

$$\varphi_n(t) = \mathbb{E}(e^{itX_n}) = \int e^{itx} d\mu_n$$

Since $h(x) = e^{itx}$ is continuous and bounded. By Theorem 3.3 we immediately have the result required.

($\Leftarrow$) Suppose $\varphi_n(t) \rightarrow \varphi(t) \forall t$. $\forall C^2$ test function h . We need to show

$$\int h d\mu_n \rightarrow \int h d\mu$$

By Parseval-Plancherel Theorem it suffices to show

$$\int \overline{\hat{h}(t)} \varphi_n(t) dt \rightarrow \int \overline{\hat{h}(t)} \varphi(t) dt$$

Note that

$$|\overline{\hat{h}(t)}\varphi_n(t)| \leq |\overline{\hat{h}(t)}| \leq \min\{\frac{C}{t^2}, M\}$$

for some M and C . Then we immediately have the result by dominated convergence theorem. $\square$

Example. Let (X_n) be independent and $\mathbb{P}(X_n = \pm 1) = \frac{1}{2}$. Does $\sum_{n=1}^N \frac{X_n}{2^n}$ converges in distribution? Let $S_n = \sum_{k=1}^n \frac{X_k}{2^k}$ then

$$\varphi_{S_k}(t) = \prod_{k=1}^n \varphi_{\frac{X_k}{2^k}}(t) = \prod_{k=1}^n \varphi\left(\frac{t}{2^k}\right) = \prod_{k=1}^n \cos\left(\frac{t}{2^k}\right) = \frac{\sin(t)}{2^n \sin\left(\frac{t}{2^n}\right)} = \frac{\sin(t)}{t}$$

as $n \rightarrow \infty$.

Theorem 3.13. If $\varphi_X = \varphi_Y$ then $F_X = F_Y$.

Proof. Let $X_n = X$, clearly $X_n \Rightarrow X$ which means $\varphi_{X_n} \rightarrow \varphi_X = \varphi_Y$

$$\begin{aligned} &\Rightarrow X_n \Rightarrow Y \\ &\Rightarrow F_{X_n}(t) \rightarrow F_Y(t) \end{aligned}$$

at all continuity points of F_Y . i.e.

$$F_X(t) = F_Y(t)$$

at all continuity points of F_Y . Let t be a discontinuity point; let $t_n \downarrow t$, where t_n are continuity points then

$$F_X(t) = \lim_{n \rightarrow \infty} F_X(t_n) = \lim_{n \rightarrow \infty} F_Y(t_n) = F_Y(t)$$

$\square$

Example. Let X and Y be independent and Cauchy-distributed. Find the distribution of $X + Y$.

The density of $X + Y$ is

$$f_{X+Y}(t) = \frac{1}{\pi^2} \int_{-\infty}^{\infty} \frac{1}{1+x^2} \cdot \frac{1}{1+(t-x)^2} dx$$

Instead, if we choose characteristic function

$$\varphi_{X+Y}(t) = e^{-2|t|} \Rightarrow \varphi_{\frac{X+Y}{2}}(t) = e^{-|t|}$$

Hence $\frac{X+Y}{2}$ is Cauchy. Then

$$\begin{aligned} F_{X+Y}(t) &= \mathbb{P}(X + Y \leq t) = \mathbb{P}\left(\frac{X + Y}{2} \leq \frac{t}{2}\right) = F_{\frac{X+Y}{2}}\left(\frac{t}{2}\right) \\ f_{X+Y}(t) &= F'_{X+Y}(t) = \frac{1}{2} f_{\frac{X+Y}{2}}\left(\frac{t}{2}\right) = \frac{2}{\pi(4+t^2)} \end{aligned}$$

Theorem 3.14. If $\mathbb{E}(|X|^m) < \infty$ then $\varphi^{(m)}$ exists, is continuous and

$$\varphi^{(m)}(t) = i^m \mathbb{E}(X^m e^{itX})$$

In particular, if $\mathbb{E}(X^2) < \infty$ then φ'' exists, is continuous and

$$\begin{aligned} \varphi'(0) &= i\mathbb{E}(X) \\ \varphi''(0) &= -\mathbb{E}(X^2) \end{aligned}$$

Proof. Induction: $m = 0$ is the definition of φ . Assume the formula true for $m = k$ then

$$\frac{\varphi^{(k)}(t+h) - \varphi^{(k)}(t)}{h} = \frac{i^k \mathbb{E}(X^k e^{i(t+h)X}) - i^k \mathbb{E}(X^k e^{itX})}{h} = i^k \mathbb{E}(X^k e^{itX} \frac{e^{ihX} - 1}{h})$$

Note that

$$\begin{aligned} \left| \frac{e^{ihX} - 1}{h} \right| &= \left| \frac{1}{h} \int_0^{hX} e^{is} ds \right| \leq |X| \\ \Rightarrow \left| X^k e^{itX} \frac{e^{ihX} - 1}{h} \right| &\leq |X^{k+1}| \end{aligned}$$

By dominated convergence theorem,

$$i^k \mathbb{E}(X^k e^{itX} \frac{e^{ihX} - 1}{h}) \rightarrow i^{k+1} \mathbb{E}(X^{k+1} e^{itX})$$

as $h \rightarrow 0$. Now let's prove $\varphi^{(m)}$ is continuous. Let $t_n \rightarrow t$ and show $\varphi(t_n) \rightarrow \varphi(t)$. Note

$$\varphi^{(m)}(t_n) = i^m \mathbb{E}(X^m e^{it_n X}) \rightarrow i^m \mathbb{E}(X^m e^{itX}) = \varphi^{(m)}(t)$$

by dominated convergence theorem. $\square$

Theorem 3.15. (Central limit theorem) Let (X_n) be iid square integrable random variables. Let $S_n = X_1 + \dots + X_n$,

$\mu = \mathbb{E}(X_n), \sigma^2 = \text{Var}(X_n)$ then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow N(0, 1)$$

i.e.

$$\frac{S_n}{n} \approx \mu + \frac{\sigma}{\sqrt{n}}N(0, 1)$$

Proof. WLOG, suppose $\mu = 0, \sigma^2 = 1$. Let's show

$$\frac{S_n}{\sqrt{n}} \Rightarrow N(0, 1)$$

note that

$$\varphi_{\frac{S_n}{\sqrt{n}}}(t) = \varphi_{\frac{X_1 + \dots + X_n}{\sqrt{n}}}(t) = \varphi\left(\frac{t}{\sqrt{n}}\right)^n$$

By Taylor expansion,

$$\varphi(s) = \varphi(0) + \varphi'(0)s + \frac{\varphi''(\xi_s)s^2}{2}$$

where $0 \leq \xi_s \leq s$. And

$$\varphi''(\xi_s) = \varphi''(0) + \varepsilon_s = -\sigma^2 + \varepsilon_s = -1 + \varepsilon_s$$

where $\varepsilon_s \rightarrow 0$. Therefore,

$$\varphi\left(\frac{t}{\sqrt{n}}\right)^n = \left(1 + (-1 + \varepsilon_{\frac{t}{\sqrt{n}}})\frac{t^2}{2n}\right)^n = \left(1 - \frac{t^2}{2n} + \varepsilon_{\frac{t}{\sqrt{n}}}\frac{t^2}{2n}\right)^n \rightarrow e^{-\frac{t^2}{2}}$$

Now, consider $Y_n = \frac{X_n - \mu}{\sigma}$ then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{X_1 + \dots + X_n - \mu n}{\sigma\sqrt{n}} = \frac{Y_1 + \dots + Y_n}{\sqrt{n}} \Rightarrow N(0, 1)$$

then the proof is completed by Theorem 3.13. □

Remark. (Weak law of large numbers)

$$\frac{X_1 + \dots + X_n}{n} \Rightarrow \mu$$

4 Martingales

(Conditional expectations) (Ω, Σ, P) , $X, \mathcal{F} \subset \Sigma$. Define

$$Y \equiv \mathbb{E}(X|\mathcal{F})$$

be a random variable that will be called the conditional expectation of X with respect to $\mathcal{F}$.

Example. Let $\Omega = \{1, 2, 3, 4\}$, $\Sigma = 2^\Omega$, $\mathbb{P}(\{1\}) = \dots = \mathbb{P}(\{4\}) = \frac{1}{4}$. $X(i) = i$. Let $\mathcal{F} = \{\{1, 2\}, \{3, 4\}, \emptyset, \Omega\}$. Since $X^{-1}(\{1\}) = \{1\}$ so X is not measurable with respect to $\mathcal{F}$. We want to find replace X with a "simpler version" Y such that Y is $\mathcal{F}$ measurable Y is as similar to X as possible or

$$\int_A X dP = \int_A Y dP, \quad \forall A \in \mathcal{F}$$

Definition 4.1. $\nu \ll \mu$ if $\mu(A) = 0 \implies \nu(A) = 0$. ν is absolutely continuous with respect to μ .

Example. $\mathbb{R}, \mu = \text{Leb}$ and $\nu =$ a measure with density f . Then

$$\nu(A) = \int_A f(x) dx = 0 \text{ if } \text{Leb}(A) = 0$$

Theorem 4.2. (Radom-Nikodym Theorem) Let μ and ν be two σ -finite measures. If $\nu \ll \mu$ then there is a measurable function f (called density) such that

$$\nu(A) = \int_A f d\mu$$

Proof. Let $(X, \mathcal{F})$ be a measurable space and let $\mu, \nu : \mathcal{F} \rightarrow [0, R]$ be two finite measures on X such that $\nu(A) = 0, \forall A \in \mathcal{F}$ such that $\mu(A) = 0$. Then $\sigma = \mu + \nu$ is a finite measure on X such that $\sigma(A) = 0$ iff $\mu(A) = 0$. Consider the linear functional $T : L^2(X, \mathcal{F}, \sigma) \rightarrow \mathbb{R}$ defined by

$$Tu = \int_X u d\mu, \quad \forall u \in L^2(X, \mathcal{F}, \sigma)$$

T is well-defined as μ is finite and dominated by σ , so that $L^2(X, \mathcal{F}, \sigma) \subseteq L^2(X, \mathcal{F}, \mu) \subseteq L^1(X, \mathcal{F}, \mu)$ it's also bounded and linear since

$$|Tu| \leq \|u\|_{L^2(X, \mathcal{F}, \sigma)} \sqrt{\sigma(X)}$$

By Riesz representation theorem, there exists unique $g \in L^2(X, \mathcal{F}, \sigma)$ such that

$$Tu = \int_X u d\mu = \int_X ug d\sigma, \quad \forall u \in L^2(X, \mathcal{F}, \sigma)$$

then $\mu(A) = \int_A g d\sigma, \forall A \in \mathcal{F}$ so that $0 < g \leq 1$ is σ, μ measurable a.e. □

Theorem 4.3. (Definition of conditional expectation) Let X be an integrable random variables on a probability space (Ω, Σ, P) , and let $\mathcal{F} \subset \Sigma$ be a sub- σ -algebra. Then there exists a random variable Y such that the followings are satisfied

1. Y is integrable.
2. Y is $\mathcal{F}$ -measurable.
- 3.

$$\int_A X dP = \int_A Y dP, \quad \forall A \in \mathcal{F}$$

Y is called the conditional expectation of X with respect to $\mathcal{F}$ and is denoted by $\mathbb{E}(X|\mathcal{F})$. If $\tilde{Y}$ is another random variables with these properties then $Y = \tilde{Y}$ a.s.

Proof. Assume $X \geq 0$ a.s. Consider $(\Omega, \mathcal{F})$. Consider two measure P and Q where

$$Q(A) = \int_A X dP, \quad Q \ll P$$

Then by Radom-Nikodym Theorem, there exists some function Y such that

$$Q(A) = \int_A Y dP$$

Now, to show the uniqueness. Suppose $\tilde{Y}$ satisfies the three properties and $\mathbb{P}(Y \neq \tilde{Y}) > 0$ note that

$$\{Y \neq \tilde{Y}\} = \left(\bigcup_{n=1}^{\infty} \{Y - \tilde{Y} > \frac{1}{n}\} \right) \cup \left(\bigcup_{n=1}^{\infty} \{Y - \tilde{Y} < -\frac{1}{n}\} \right)$$

$\exists n$ such that $\mathbb{P}(Y - \tilde{Y} > \frac{1}{n}) = p > 0$. WLOG, since A and $\tilde{A}$ are $\mathcal{F}$ -measurable

$$0 = \int_{\{Y - \tilde{Y} > \frac{1}{n}\}} (X - \tilde{X}) dP = \int_{\{Y - \tilde{Y} > \frac{1}{n}\}} (Y - \tilde{Y}) dP \geq \frac{1}{n} \cdot p > 0$$

□

Theorem 4.4. (Properties of conditional expectation)

- (a) $\mathbb{E}(\mathbb{E}(X|\mathcal{F})) = \mathbb{E}(X)$.
- (b) If X is $\mathcal{F}$ -measurable, then $\mathbb{E}(X|\mathcal{F}) = X$.
- (c) (Linearity) $\mathbb{E}(\lambda_1 X_1 + \lambda_2 X_2|\mathcal{F}) = \lambda_1 \mathbb{E}(X_1|\mathcal{F}) + \lambda_2 \mathbb{E}(X_2|\mathcal{F})$ for $\lambda_1, \lambda_2 \in \mathbb{C}$.
- (d) If $X \geq 0$ a.s. then $\mathbb{E}(X|\mathcal{F}) \geq 0$ a.s.
- (e) Conditional monotone. If $0 \leq X_n \uparrow X$ a.s. then $\mathbb{E}(X_n|\mathcal{F}) \uparrow \mathbb{E}(X|\mathcal{F})$ a.s.
- (f) $\mathbb{E}(ZX|\mathcal{F}) = Z\mathbb{E}(X|\mathcal{F})$ if Z is $\mathcal{F}$ -measurable.
- (g) $\mathbb{E}(X|\mathcal{F}) = \mathbb{E}(X)$ if X is independent of $\mathcal{F}$.
- (h) (Tower property) Consider $\sigma \subseteq \mathcal{F} \subseteq \Sigma$ then

$$\mathbb{E}(\mathbb{E}(X|\mathcal{F})|\sigma) = \mathbb{E}(X|\sigma)$$

- (i) (Conditional Jensen inequality) If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex then

$$\mathbb{E}(\varphi(X)|\mathcal{F}) \geq \varphi(\mathbb{E}(X|\mathcal{F}))$$

Examples.

- (1) Let (X_n) be i.i.d., consider $S_n = X_1 + \dots + X_n$ if $m < n$ then

$$\begin{aligned} \mathbb{E}(S_n|\sigma(X_1, \dots, X_m) = \sigma_m) &= \mathbb{E}(X_1 + \dots + X_n|\sigma_m) \\ &= \mathbb{E}(X_1 + \dots + X_m|\sigma_m) + \mathbb{E}(X_{m+1} + \dots + X_n|\sigma_m) \\ &= X_1 + \dots + X_m + \mathbb{E}(X_{m+1} + \dots + X_n) \\ &= S_m + \mu(n - m) \end{aligned}$$

where $\mu = \mathbb{E}(X_i)$, $\forall i$.

- (2) Toss q -coin with outcome X , then toss p_X -coin with outcome Y . Then

$$\mathbb{E}(Y|\sigma(X)) = p_X$$

Definition 4.5. A growing sequence $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ of σ -algebra is called a filtration. A random process (X_n) is adapted to a filtration $(\mathcal{F}_n)$ if X_n is $\mathcal{F}_n$ -measurable $\forall n$. $\mathcal{F}_n = \sigma(x_0, \dots, x_n)$ is called the natural filtration of (X_n) . (X_n) is always adapted to its natural filtration.

Definition 4.6. A process (X_n) is a martingale with respect to a filtration $\mathcal{F}_n$ if

- (a) all X_n are integrable.
- (b) (X_n) is adapted to $(\mathcal{F}_n)$.
- (c) $\mathbb{E}(X_{n+1}|\mathcal{F}) = X_n$, $\forall n$.

Definition 4.7. process (X_n) is a submartingale with respect to a filtration $\mathcal{F}_n$ if

- (a) all X_n are integrable.
- (b) (X_n) is adapted to $(\mathcal{F}_n)$.
- (c) $\mathbb{E}(X_{n+1}|\mathcal{F}) \geq X_n$, $\forall n$.

Remark. If $m < n$, then

$$\mathbb{E}(X_n|\mathcal{F}_m) = X_m$$

Since by tower property,

$$\mathbb{E}(\mathbb{E}(X_n|\mathcal{F}_{n-1})|\mathcal{F}_m) = \mathbb{E}(X_{n-1}|\mathcal{F}_m)$$

induction gives

$$\mathbb{E}(X_n|\mathcal{F}_m) = X_m$$

Remark. $\mathbb{E}(X_n) = \mathbb{E}(X_0)$.

Given a martingale (X_n) , we want to investigate the gambling strategy, to determine time T when you end the game.

Definition 4.8. $T : \Omega \rightarrow \{0, \dots\} \cup \mathbb{N} \cup \{+\infty\}$ is a stopping time if

$$\{T = n\} \in \mathcal{F}_n, \forall n$$

Example. Given $\mathcal{F}_n = \{\text{weather in days } 1, \dots, n\}$ then

$T_1 = \text{first raining day}$

$T_2 = \text{second raining day}$

$T_3 = \text{last raining day before a sunny day}$

then T_1, T_2 are stopping times but T_3 is not.

Example. (X_n) —adapted to $(\mathcal{F}_n)$ then

$$T = \inf\{n : X_n > 50\}$$

is the stopping time.

Example. $S_n = \mathcal{E}_1 + \dots + \mathcal{E}_n$ where $\mathbb{P}(\mathcal{E}_n = \pm 1) = \frac{1}{2}$ and $\mathcal{F}_n = \sigma(\mathcal{E}_1, \dots, \mathcal{E}_n)$. Define $T = \inf\{n : S_n = a \text{ or } S_n = -b\}$ then

$$\{T = n\} = \{S_1, \dots, S_{n-1} \notin \{a, -b\}, S_n \in \{a, -b\}\} \in \mathcal{F}_n$$

is a stopping time.

Lemma 4.9. T is the stopping time if and only if

$$\{T \leq n\} \in \mathcal{F}_n, \forall n$$

Proof. Note that

$$\begin{aligned} \{T \leq n\} &= \{T = 0\} \cup \{T = 1\} \cup \dots \cup \{T = n\} \in \mathcal{F}_n \\ \{T = n\} &= \{T \leq n\} \cap \{T \leq n\}^c = \{T \leq n\} \cap \{T \leq n-1\}^c \in \mathcal{F}_n \end{aligned}$$

□

Theorem 4.10. Let (X_n) be a martingale and T a.s. finite stopping time. Then $(X_{n \wedge T})$ is also a martingale.

Proof. By definition,

$$X_{n \wedge T} = \sum_{i=0}^{n-1} \mathbb{1}_{\{T \geq i\}} X_i + X_n \mathbb{1}_{\{T \geq n\}} \in \mathcal{F}_n$$

and note that

$$\begin{aligned} \mathbb{E}(X_{n \wedge T} | \mathcal{F}_{n-1}) &= \mathbb{E}\left(\sum_{i=0}^{n-1} \mathbb{1}_{\{T \geq i\}} X_i + X_n \mathbb{1}_{\{T \geq n\}} \mid \mathcal{F}_{n-1}\right) \\ &= \sum_{i=0}^{n-1} \mathbb{1}_{\{T \geq i\}} X_i + \mathbb{1}_{\{T \geq n\}} \mathbb{E}(X_n | \mathcal{F}_{n-1}) \\ &= \sum_{i=0}^{n-2} \mathbb{1}_{\{T \geq i\}} X_i + \mathbb{1}_{\{T \geq n-1\}} X_{n-1} = X_{(n-1) \wedge T} \end{aligned}$$

□

Theorem 4.11. (Optional Stopping time) Let (X_n) be a martingale and T is an a.s. finite stopping time with respect to filtration $(\mathcal{F}_n)$. Then $\mathbb{E}(X_T) = \mathbb{E}(X_0)$ in each of the following scenarios:

- (1) T is a.s. bounded.
- (2) $\mathbb{E}(T) < \infty$ and $|X_{n+1} - X_n|$ are a.s. bounded.
- (3) $|X_n|$ is a.s. bounded.

Proof.

- (1) If $T \leq N$ a.s. $X_{N \wedge T} = X_T$ since (X_n) is a martingale so

$$\mathbb{E}(X_0) = \mathbb{E}(X_{N \wedge T}) = \mathbb{E}(X_T)$$

- (2) Since $X_{n \wedge T(\omega)} \rightarrow X_{T(\omega)}$ a.s. as $n \rightarrow \infty$,

$$|X_{n \wedge T} - X_0| \leq \sum_{i=1}^{n \wedge T} |X_i - X_{i-1}| \leq cT$$

so

$$|X_{n \wedge T}| \leq |X_0| + cT < \infty$$

then by dominated convergence theorem,

$$\mathbb{E}(X_{n \wedge T}) \rightarrow \mathbb{E}(X_T) = \mathbb{E}(X_0)$$

- (3) Since $X_{n \wedge T(\omega)} \rightarrow X_{T(\omega)}$ a.s. as $n \rightarrow \infty$, by dominated convergence theorem since $|X_n|$ is a.s. bounded so

$$\mathbb{E}(X_{n \wedge T}) \rightarrow \mathbb{E}(X_T) = \mathbb{E}(X_0)$$

□

Examples.

- (1) Random walk $S_n = \mathcal{E}_1 + \dots + \mathcal{E}_n$ and $T = \inf\{n : S_n \in \{a, -b\}\}$. Then $\mathbb{E}(S_T) = \mathbb{E}(S_0) = 0$ and

$$\begin{aligned}\mathbb{E}(S_T) &= a\mathbb{P}(S_T = a) - b\mathbb{P}(S_T = -b) \\ \implies \mathbb{P}(S_T = a) &= \frac{b}{b+a}, \quad \mathbb{P}(S_T = -b) = \frac{a}{a+b}\end{aligned}$$

We want to justify $T < \infty$ a.s. note that $\forall n$

$$\{T = \infty\} \subseteq \{|S_n| \leq \max\{|a|, |b|\}\} = \{|\frac{S_n}{\sqrt{n}}| \leq \frac{\max\{\dots\}}{\sqrt{n}}\}$$

$$\mathbb{P}(T = \infty) \leq \lim_{n \rightarrow \infty} \mathbb{P}(|\frac{S_n}{\sqrt{n}}| \leq \frac{\max\{\dots\}}{\sqrt{n}}) = 0$$

by central limit theorem. Then optional stopping time theorem is applicable.

- (2) $X_n = S_n^2 - n$ is a martingale. Since

$$\begin{aligned}\mathbb{E}(X_{n+1}|\mathcal{F}_n) &= \mathbb{E}(S_{n+1}^2 - (n+1)|\mathcal{F}_n) = \mathbb{E}(S_n^2 + 2S_n\mathcal{E}_{n+1} + \mathcal{E}_{n+1}^2 - n - 1|\mathcal{F}_n) \\ &= S_n^2 + 2S_n \cdot 0 + 1 - n - 1 = X_n\end{aligned}$$

By the optional stopping time theorem $\mathbb{E}(X_T) = \mathbb{E}(X_0)$ then

$$\mathbb{E}(S_T^2) - \mathbb{E}(T) = \mathbb{E}(X_T) = \mathbb{E}(X_0) = 0$$

$$\mathbb{E}(T) = \mathbb{E}(S_T^2) = a^2 \frac{b}{a+b} + b^2 \frac{a}{a+b} = ab$$

In order to apply the optional stopping time theorem, we know that

$$S_{n \wedge T}^2 - (n \wedge T) \rightarrow S_T^2 - T \quad a.s.$$

we want to show that

$$\mathbb{E}(S_{n \wedge T}^2 - (n \wedge T)) \rightarrow \mathbb{E}(S_T^2 - T) \quad a.s.$$

note that by dominated convergence theorem we have

$$\mathbb{E}(S_{n \wedge T}^2) \rightarrow \mathbb{E}(S_T^2)$$

and in fact by monotone converging theorem,

$$\mathbb{E}(n \wedge T) \rightarrow \mathbb{E}(T)$$

therefore,

$$\mathbb{E}(S_{n \wedge T}^2 - (n \wedge T)) \rightarrow \mathbb{E}(S_T^2 - T) \quad a.s.$$

- (3) Alice and Bob are rolling a die and wait to see the sequence 456 and 666 to end the game. We want to compute $\mathbb{E}(T_{666})$ and $\mathbb{E}(T_{456})$ respectively. Let $(\mathcal{E}_i)$ be the independent die rolls. Denote $M_n = W_n - n$ where W_n is the money won by all gamblers by time n and n is the money invested by all gamblers by time n . $\mathbb{E}(M_{T_{666}}) = \mathbb{E}(M_0) = 0$ then

$$\mathbb{E}(M_{T_{666}}) = \mathbb{E}(T_{666}) = 6^3 + 6^2 + 6$$

Similarly,

$$\mathbb{E}(M_{T_{456}}) = \mathbb{E}(T_{456}) = 6^3$$

Theorem 4.12. (Doob's submartingale inequality) Let (X_n) be a non-negative submartingale. Then $\forall c > 0$,

$$c\mathbb{P}(E = \max_{1 \leq n \leq N} X_n \geq c) \leq \mathbb{E}(X_N)$$

Proof. Let $T = \inf\{n : X_n \geq c\} \wedge N$ be the stopping time. We claim that

$$T \leq N \implies \mathbb{E}(X_T) \leq \mathbb{E}(X_N)$$

Note that since (X_n) is submartingale

$$\begin{aligned}\mathbb{E}(X_T) &= \mathbb{E}(\sum_{i=0}^N \mathbb{1}_{\{T=i\}} X_i) = \sum_{i=0}^N \int_{\{T=i\} \in \mathcal{F}_i} X_i dP \leq \sum_{i=0}^N \int_{\{T=i\} \in \mathcal{F}_i} \mathbb{E}(X_N | \mathcal{F}_i) dP \\ &= \sum_{i=0}^N \int_{\{T=i\}} X_N dP = \sum_{i=0}^N \mathbb{E}(X_N \mathbb{1}_{\{T=i\}}) = \mathbb{E}(X_N \sum_{i=0}^N \mathbb{1}_{\{T=i\}}) = \mathbb{E}(X_N)\end{aligned}$$

Let $\mathbb{E}(X_T \mathbb{1}_E) + \mathbb{E}(X_T \mathbb{1}_{\Omega \setminus E}) = \mathbb{E}(X_T)$ and $\mathbb{E}(X_N \mathbb{1}_E) + \mathbb{E}(X_N \mathbb{1}_{\Omega \setminus E}) = \mathbb{E}(X_N)$. Then $\mathbb{E}(X_T \mathbb{1}_{\Omega \setminus E}) = \mathbb{E}(X_N \mathbb{1}_{\Omega \setminus E})$. Therefore we have

$$\begin{aligned}\mathbb{E}(X_T \mathbb{1}_E) &\leq \mathbb{E}(X_N \mathbb{1}_E) \\ \mathbb{E}(X_N) &\geq \mathbb{E}(X_N \mathbb{1}_E) \geq \mathbb{E}(X_T \mathbb{1}_E) \geq c\mathbb{P}(E)\end{aligned}$$

□

Theorem 4.13. (Kolmogorov's inequality) Let (X_n) be independent, square integrable with mean zero. Then $\forall c > 0$,

$$c^2 \mathbb{P}(\max_{1 \leq n \leq N} |\sum_{i=1}^n X_i| > c) \leq \sum_{i=1}^N \mathbb{E}(X_i^2)$$

Proof. LHS of the inequality is equivalent to

$$c^2 \mathbb{P}(\max_{1 \leq n \leq N} |\sum_{i=1}^n X_i|^2 > c^2)$$

if $(\sum_{i=1}^n X_i)^2$ are in fact submartingale then by Doob's submartingale inequality we have

$$c^2 \mathbb{P}(\max_{1 \leq n \leq N} |\sum_{i=1}^n X_i|^2 > c^2) \leq \mathbb{E}(\sum_{i=1}^N X_i^2) = \sum_{i=1}^N \mathbb{E}(X_i^2)$$

so it suffices to show that $(\sum_{i=1}^n X_i)^2$ are in fact submartingale. Let $S_n = \sum_{i=1}^n X_i$ then note that

$$\mathbb{E}(S_n | \mathcal{F}_{n-1}) = \mathbb{E}(S_{n-1} + X_n | \mathcal{F}_{n-1}) = S_{n-1}$$

so S_n is a martingale. It suffices to show that S_n^2 is a submartingale. Note that by convexity, $\mathbb{E}(\varphi(S)|\mathcal{F}) \geq \varphi(\mathbb{E}(S|\mathcal{F}))$ so S_n^2 is a submartingale. $\square$

Theorem 4.14. (Kolmogorov's Theorem) Let (X_n) be independent square-integrable with mean zero then if $\sum_{n=1}^{\infty} \mathbb{E}(X_n^2) < \infty$ then $\sum_{i=1}^{\infty} X_n < \infty$ a.s.

Proof. We want $\mathbb{P}((\sum_{i=1}^n X_i)_{n \in \mathbb{N}} \text{ is Cauchy}) = 1$. i.e.

$$\mathbb{P}(\forall k, \exists N, \forall n \geq N | \sum_{i=N+1}^n X_i| < \frac{1}{k}) = 1$$

i.e.

$$\mathbb{P}(\bigcup_{n \in \mathbb{N}} \bigcap_{n \geq N} \{ | \sum_{i=N+1}^n X_i| < \frac{1}{k} \}) = 1, \quad \forall k$$

i.e.

$$\mathbb{P}(\bigcap_{N=1}^{\infty} \bigcup_{n \geq N} \{ | \sum_{i=N+1}^n X_i| \geq \frac{1}{k} \}) = 0, \quad \forall k$$

note that

$$\bigcup_{n=N+1}^M \{ | \sum_{i=N+1}^n X_i| \geq \frac{1}{k} \} = \{ \max_{N+1 \leq n \leq M} | \sum_{i=N+1}^n X_i| \geq \frac{1}{k} \}$$

by Kolmogorov's inequality we have

$$\begin{aligned} \mathbb{P}(\{ \max_{N+1 \leq n \leq M} | \sum_{i=N+1}^n X_i| \geq \frac{1}{k} \}) &\leq k^2 \sum_{i=N+1}^M \mathbb{E}(X_i^2) \\ \Rightarrow \mathbb{P}(\bigcup_{n \geq N+1} \{ | \sum_{i=N+1}^n X_i| \geq \frac{1}{k} \}) &\leq k^2 \sum_{i=N+1}^{\infty} \mathbb{E}(X_i^2) \\ \Rightarrow \mathbb{P}(\bigcap_{N=1}^{\infty} \bigcup_{n \geq N+1} \{ | \sum_{i=N+1}^n X_i| \geq \frac{1}{k} \}) &\leq \lim_{N \rightarrow \infty} k^2 \sum_{i=N+1}^{\infty} \mathbb{E}(X_i^2) = 0 \end{aligned}$$

$\square$

Example. Given $\mathbb{P}(\mathcal{E}_n = \pm 1) = \frac{1}{2}$ then $\sum_{n=1}^{\infty} \frac{\mathcal{E}_n}{n} < \infty$ a.s. since $\sum_{n=1}^{\infty} \mathbb{E}((\frac{\mathcal{E}_n}{n})^2) < \infty$.

Definition 4.15. (x_n) is Cauchy iff $\forall \varepsilon, \exists N$ such that $\forall n \geq N, |x_n - x_N| < \varepsilon$.

Lemma 4.16. (Cesaro) If $a_n \rightarrow a$ then

$$\frac{a_1 + \cdots + a_n}{n} \rightarrow a$$

Lemma 4.17. (Kronecker) If $\sum_{n=1}^{\infty} \frac{a_n}{n} < \infty$ then

$$\frac{a_1 + \cdots + a_n}{n} \rightarrow 0$$

Theorem 4.18. (SLLN for square-integrable r.v.) Let (X_n) be i.i.d. square integrable random variables with mean 0. Then

$$\frac{X_1 + \cdots + X_n}{n} \rightarrow 0 \quad a.s.$$

Proof. By Kronecker, it's enough to show $\sum_{n=1}^{\infty} \frac{X_n}{n} < \infty$ a.s. By Kolmogorov, it's enough to check that $\sum_{n=1}^{\infty} \frac{\mathbb{E}(X_n^2)}{n^2} < \infty$

∞ . Note that

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}(X_n^2)}{n^2} = \mathbb{E}(X_1^2) \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

□

Theorem 4.19. (Truncation lemma) Let (X_n) be i.i.d. integrable random variables with $\mathbb{E}(X_n) = \mu$. Let

$$Y_n = X_n \mathbb{1}_{\{|X_n| \leq n\}}$$

then

- (1) $X_n = Y_n$ eventually.
- (2) $\mathbb{E}(Y_n) \rightarrow \mu$.
- (3) $\sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} < \infty$.

Proof.

- (1) We want to show $\mathbb{P}(X_n \neq Y_n \text{ i.o.}) = 0$ then it suffices to show that

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) < \infty$$

then

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_n| > n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n)$$

by Monotone converging theorem,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n) = \sum_{n=1}^{\infty} \mathbb{E}(\mathbb{1}_{\{|X_1| > n\}}) = \mathbb{E}\left(\sum_{n=1}^{\infty} \mathbb{1}_{\{|X_1| > n\}}\right)$$

then

$$\mathbb{E}\left(\sum_{n=1}^{\infty} \mathbb{1}_{\{|X_1| > n\}}\right) \leq \mathbb{E}(|X_1|) < \infty$$

- (2) $\mathbb{E}(Y_n) = \mathbb{E}(X_n \mathbb{1}_{\{|X_n| \leq n\}}) = \mathbb{E}(X_1 \mathbb{1}_{\{|X_1| \leq n\}})$. Note that $\mathbb{E}(X_1 \mathbb{1}_{\{|X_1| \leq n\}}) \rightarrow \mathbb{E}(X_1)$ since $|X_1 \mathbb{1}_{\{|X_1| \leq n\}}| \leq |X_1|$, $\forall n$.
- (3) By definition

$$\sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} = \sum_{n=1}^{\infty} \frac{\mathbb{E}(Y_n^2)}{n^2} - \sum_{n=1}^{\infty} \frac{(\mathbb{E}(Y_n))^2}{n^2}$$

we shall first show that the two series on the RHS in fact exists. First note that $\mathbb{E}(Y_n) \rightarrow \mu$ so $|\mathbb{E}(Y_n)| \leq c$ for some constant c so

$$\sum_{n=1}^{\infty} \frac{(\mathbb{E}(Y_n))^2}{n^2} \leq \sum_{n=1}^{\infty} \frac{c^2}{n^2} < \infty$$

Now when we look at another series,

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}(Y_n^2)}{n^2} = \sum_{n=1}^{\infty} \frac{\mathbb{E}(X_n^2 \mathbb{1}_{\{|X_n| \leq n\}})}{n^2} = \sum_{n=1}^{\infty} \frac{\mathbb{E}(X_1^2 \mathbb{1}_{\{|X_1| \leq n\}})}{n^2}$$

Then, by MCT

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}(X_1^2 \mathbb{1}_{\{|X_1| \leq n\}})}{n^2} = \mathbb{E}\left(X_1^2 \sum_{n=1}^{\infty} \frac{\mathbb{1}_{\{|X_1| \leq n\}}}{n^2}\right) = \mathbb{E}\left(X_1^2 \sum_{n \geq |X_1|} \frac{1}{n^2}\right)$$

Note that $\sum_{n \geq m} \frac{1}{n^2} \leq \frac{2}{m}$ so

$$\mathbb{E}\left(X_1^2 \sum_{n \geq |X_1|} \frac{1}{n^2}\right) \leq \mathbb{E}\left(X_1^2 \frac{2}{|X_1|}\right) = 2\mathbb{E}(|X_1|) < \infty$$

□

Theorem 4.20. (SLLN) Let (X_n) be i.i.d integrable random variables with $\mathbb{E}(X_n) = \mu$ then

$$\frac{X_1 + \cdots + X_n}{n} \rightarrow \mu$$

Proof. Define Y_n as in Truncation lemma, since $X_n = Y_n$ eventually so it's enough to show

$$\frac{Y_1 + \cdots + Y_n}{n} \rightarrow \mu$$

note that

$$\frac{Y_1 + \cdots + Y_n}{n} = \frac{(Y_1 - \mathbb{E}(Y_1)) + \cdots + (Y_n - \mathbb{E}(Y_n))}{n} + \frac{\mathbb{E}(Y_1) + \cdots + \mathbb{E}(Y_n)}{n}$$

Since $Y_n - \mathbb{E}(Y_n) = 0$ then it suffices to check that $Y_n - \mathbb{E}(Y_n)$ is square integrable. Note that

$$\sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} < \infty$$

which completes the proof since

$$\frac{\mathbb{E}(Y_1) + \cdots + \mathbb{E}(Y_n)}{n} \rightarrow \mu$$

□

Definition 4.21. (C_n) is a previsible process if C_n is $\mathcal{F}_{n-1}$ measurable $\forall n$.

Definition 4.22. Let (X_n) be a martingale and (C_n) be a previsible process. Then $(C \circ X)_n = \sum_{i=1}^n C_i(X_i - X_{i-1})$ is the martingale transform of (X_n) by (C_n) .

Example.

$$(X_n) \rightsquigarrow (B_t)$$

then

$$\sum C_i(X_i - X_{i-1}) \rightsquigarrow \int_0^t H_s dB_s$$

Theorem 4.23. If (C_n) is bounded then $(C \circ X)_n$ is a martingale.

Proof. Obviously, $(C \circ X)_n$ is $\mathcal{F}_n$ -measurable and integrable. Then note that

$$\begin{aligned} \mathbb{E}((C \circ X)_n | \mathcal{F}_{n-1}) &= \mathbb{E}((C \circ X)_{n-1} + C_n(X_n - X_{n-1}) | \mathcal{F}_{n-1}) \\ &= (C \circ X)_{n-1} + C_n(\mathbb{E}(X_n | \mathcal{F}_{n-1}) - X_{n-1}) = (C \circ X)_{n-1} \end{aligned}$$

since (X_n) is a martingale. □

Definition 4.24. Let (X_n) be a random sequence, let $U_N[a, b]$ is the largest number $K \in \mathbb{N}$ such that there are $0 \leq s_1 < t_1 < \cdots < s_K < t_K < N$ such that

$$X_{s_i} < a \quad \forall i \text{ and } X_{t_i} > b \quad \forall i$$

Theorem 4.25. (Doob's Upcrossing Lemma) Let (X_n) be a martingale then $(b - a)\mathbb{E}(U_N[a, b]) \leq \mathbb{E}(X_N - a)_-$.

Proof. Let $K = U_N[a, b]$ and $0 \leq s_1 < t_1 < \cdots < s_K < t_K \leq N$. $C_0 = \cdots, C_{s_1} = 0$ and $C_{s_1+1} = \cdots = C_{t_1} = 1$ and etc. Let $Y_n = \sum_{i=1}^n C_i(X_i - X_{i-1})$ which is a martingale so in particular $\mathbb{E}(Y_N) = 0$. Then

$$\begin{aligned} Y_N &\geq \sum_{i=1}^K (X_{t_i} - X_{s_i}) - (X_N - a)_- \\ \mathbb{E}(Y_N) = 0 &\implies (b - a)\mathbb{E}(U_N[a, b]) - \mathbb{E}(X_N - a)_- \leq 0 \end{aligned}$$

□

Theorem 4.26. (Martingale Convergence Theorem) Let (X_n) be a martingale bounded in L^1 . Then there is a random variable X on the same probability space such that $X_n \rightarrow X$ a.s.

Proof. Consider

$$\{\liminf_{n \rightarrow \infty} X_n \neq \limsup_{n \rightarrow \infty} X_n\} = \bigcup_{a < b; a, b \in \mathbb{Q}} \{U[a, b] = \infty\}$$

then by Doob's Upcrossing Lemma

$$\begin{aligned} (b - a)\mathbb{E}(U[a, b]) &= (b - a) \lim_{N \rightarrow \infty} \mathbb{E}(U_N[a, b]) \leq \limsup_{N \rightarrow \infty} \mathbb{E}(X_N - a)_- \leq \limsup_{N \rightarrow \infty} (\mathbb{E}|X_N| + |a|) < \infty \\ &\implies U(a, b) < \infty \text{ a.s.} \end{aligned}$$

so

$$\{\liminf_{n \rightarrow \infty} X_n \neq \limsup_{n \rightarrow \infty} X_n\} = \bigcup_{a < b; a, b \in \mathbb{Q}} \{U[a, b] = \infty\}$$

has probability zero. Then we have $\liminf_{n \rightarrow \infty} X_n = \limsup_{n \rightarrow \infty} X_n$ a.s. Now we want to show $X := \liminf_{n \rightarrow \infty} X_n < \infty$ a.s. Then

$$\mathbb{E}|X| = \mathbb{E}(\liminf_{n \rightarrow \infty} |X_n|) \leq \liminf_{n \rightarrow \infty} \mathbb{E}|X_n| \leq c$$

so $|X| < \infty$ a.s. so $X < \infty$ a.s. □